Midterm 2

Time Limit: 50 minutes

- **Every non-trivial calculation and claim in the exam must be properly justified.**
- Your work must be well organized and coherent. Pay attention to notation, definitions of treatment and outcome variables, units of measurement, and interpretation.
- **You must answer each question inside the designated box**; answers outside will not be considered.
- The use of calculators is allowed.
- Digital devices such as mobile phones, tablets, and laptops, as well as internet access, are forbidden.

The following questions are about fictional studies of education and civic participation. The exam focuses on causal designs, study assumptions, and interpretation of linear regression output. No question requires writing R code.

1. (2.5 points) A regional government wants to evaluate whether smaller seminar groups improve student outcomes in a required civic-education course. Among students enrolled in the course, some are randomly assigned to a small seminar group and others remain in a regular seminar group. The outcome is the final exam score, measured from 0 to 100.

Group	Number of students	Mean final score	Standard deviation
Small seminar ($T = 1$)	160	74	12
Regular seminar ($T = 0$)	180	70	14

(a) Identify the treatment variable, treatment group, control group, and outcome variable.

(b) Define $Y_i(1)$, $Y_i(0)$, and the individual treatment effect TE_i in this study.

(c) Compute the difference-in-means estimator. Under the study design, what is its causal interpretation?

(d) Name one possible violation of SUTVA in this study and explain why it matters.

2. (2.5 points) The same region later introduces free after-school tutoring in the North district. The South district does not receive the program. The outcome is the average absenteeism rate, measured as missed school days per 100 student-days.

District	2025 absenteeism	2026 absenteeism
North district, tutoring program	9.0	6.5
South district, no program	7.0	6.2

(a) Identify the treatment group, control group, before period, after period, and outcome variable.

(b) Compute the before-and-after estimator for the North district. Why might this be a biased causal estimate?

(c) Compute the difference-in-differences estimator and interpret it.

(d) State the key assumption behind difference-in-differences and give one possible violation in this context.

3. (2.5 points) A nonprofit evaluates a voluntary digital-literacy workshop. Participants choose whether to attend. The outcome is a civic-information score from 0 to 100. Before the workshop, residents are classified by prior interest in politics.

Prior interest stratum	Workshop mean	No-workshop mean	Estimated effect	Population share
High prior interest	78	70	8	0.40
Low prior interest	58	54	4	0.60

The unadjusted comparison in the full sample is:

$$\bar{Y}_{\text{workshop}} - \bar{Y}_{\text{no workshop}} = 10.5.$$

(a) Why is the unadjusted difference of 10.5 points not automatically a credible causal estimate?

(b) Compute the subclassification estimate using the population shares as weights.

(c) Compare the unadjusted estimate to the subclassification estimate. What does the difference suggest?

(d) What assumption is still needed for the subclassification estimate to have a causal interpretation?

4. (2.5 points) A researcher studies political knowledge among 300 residents. The dependent variable *knowledge* is a score from 0 to 100. The researcher estimates the following regression:

$$\widehat{knowledge} = 35 + 4\,workshop + 3.2\,news + 6\,college, \qquad R^2 = 0.42.$$

The variables are:

- $workshop = 1$ if the resident attended a civic workshop and 0 otherwise;
- $news$ is the number of hours per week the resident follows political news;
- $college = 1$ if the resident has a university degree and 0 otherwise.

In the estimation sample, $news$ ranges from 0 to 8 hours per week.

(a) Interpret the coefficients on $workshop$ and $news$ using the ceteris paribus idea.

(b) Compute the predicted knowledge score for a resident who attended the workshop, follows political news 5 hours per week, and does not have a university degree.

(c) Suppose this resident's observed score is 61. Compute and interpret the residual.

(d) Interpret $R^2 = 0.42$. Would you use this model to predict a resident who follows political news 15 hours per week? Explain.

(e) Can the coefficient on *workshop* automatically be interpreted as the causal effect of attending the workshop? Why or why not?